Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Crime Busters | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which solid turns brown iodine colorless?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Cornstarch

B. Table salt

C. Gypsum

D. Vitamin C

Correct: D

2. (2 points) Which listed metal does not react with hydrochloric acid?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Iron

B. Copper

C. Zinc

D. Magnesium

Correct: B

3. (2 points) A human hair usually has a medullary index of:

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair’s width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

A. Greater than one-half

B. Exactly one

C. Zero always

D. Less than about one-third

Correct: D

4. (2 points) Which plastics float in water?

Unit: Recyclable plastics.
Worked-case data: A plastic fragment sinks in water, floats in a 1.10 g/cm^3 salt solution, and burns with a sooty flame. Identify it.
Separate investigation observations:
Water: Sinks.
1.10 g/cm^3 solution: Sinks.
1.25 g/cm^3 solution: Floats.
Appearance: Very clear, rigid, and shatter-resistant.

A. All plastics

B. PP, LDPE, and HDPE

C. PVC and PETE

D. PS and PMMA

Correct: B

5. (2 points) Why is the starting line drawn in pencil?

Unit: Paper chromatography.
Worked-case data: Ransom-note ink separates into a blue spot at 2.4 cm and a yellow spot at 4.8 cm, with the solvent front at 6.0 cm. Suspect A's pen gives Rf values 0.40 and 0.80; suspect B's gives 0.40 and 0.65. Which pen matches?
Separate investigation observations:
Note ink: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 1 pen: Two bands: Rf 0.45, 0.70.
Suspect 2 pen: Three bands: Rf 0.15, 0.45, 0.70.
Suspect 3 pen: Three bands: Rf 0.20, 0.50, 0.90.

A. Pencil speeds up the solvent

B. Ink is required for Rf

C. Pencil changes Rf values

D. Pencil graphite does not dissolve and separate

Correct: D

6. (2 points) Which fingerprint pattern has no delta?

Unit: Fingerprints, DNA, prints, soil, and spatter.
Worked-case data: A scene DNA sample shows peaks at 12 and 15 repeats at location 1, and 8 and 8 at location 2. Suspect X: 12/15 and 8/10. Suspect Y: 12/15 and 8/8. Which suspect matches?
Separate investigation observations:
Scene print: A whorl with two deltas.
Suspect A: Right index finger: loop.
Suspect B: Right index finger: whorl with matching ridge endings.
Suspect C: Right index finger: arch.

A. Double loop

B. Arch

C. Loop

D. Whorl

Correct: B

7. (2 points) An analysis must explain the culprit and also:

Unit: Writing the crime scene analysis.
Worked-case data: Evidence: the scene powder is baking soda; suspect A has baking soda, B has cornstarch, C has baking soda. Note ink matches only C's pen. A dog hair was found; A and C own dogs, B owns a cat. Who is the culprit?
Separate investigation observations:
Powder: Calcium carbonate: found with suspects 1 and 3.
Plastic: HDPE fragment: suspects 1, 2, and 3 all use HDPE jugs.
Chromatography: Note ink matches only suspect 3's pen.
Fingerprint: Scene loop matches suspect 3's minutiae.

A. Only the strongest evidence

B. Each suspect's alibi only

C. The price of each test

D. Why the other suspects were not chosen

Correct: D

8. (2 points) Which solid fizzes in plain water without added acid?

Unit: Identifying white powders.
Worked-case data: An unknown white powder is insoluble in water, fizzes strongly with HCl, and leaves iodine yellow-brown. Identify it.
Separate investigation observations:
Appearance: Fine, pure white powder that squeaks when pressed.
Water: Does not dissolve; forms a cloudy suspension.
HCl: No fizzing.
Iodine: Turns blue-black.

A. Baking soda

B. Calcium carbonate

C. Sugar

D. Alka-Seltzer

Correct: D

9. (2 points) Which liquid is basic with a strong, pungent odor?

Unit: Metals and liquids.
Worked-case data: A silvery metal is not magnetic, reacts vigorously with HCl, and has a density of about 1.7 g/cm^3. Identify it.
Separate investigation observations:
Liquid color: Clear and colorless.
Odor: Little odor.
pH paper: About pH 6, close to neutral.
With yeast: Vigorous foaming and bubbling.

A. Lemon juice

B. Household ammonia

C. Vinegar

D. Hydrogen peroxide

Correct: B

10. (2 points) A fiber burns quickly, glows afterward, and leaves soft gray ash. It is:

Unit: Hair and fibers.
Worked-case data: A hair has a continuous medulla about 60% of the hair's width and scales shaped like petals. What is its source?
Separate investigation observations:
Hair A: Medulla fragmented; index about 0.2.
Hair B: Wide, continuous ladder-like medulla; flattened scales.
Fiber: Burns quickly with afterglow and soft gray ash.

A. Animal (such as wool)

B. Synthetic (such as nylon)

C. Mineral

D. Vegetable (such as cotton)

Correct: D

11. (2 points) Identify the powder from the supplied reference comparison and distinguish observation from inference.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: Sodium bicarbonate is the supported identification: solubility and acid-gas observations match that supplied reference. The identity is an inference from the observations. (2 points).

12. (2 points) Why is gas production with acid alone insufficient to uniquely identify this powder?

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: More than one material can evolve gas with acid; compare a set of diagnostic results against appropriate references. (2 points).

13. (2 points) Calculate the Rf value for the first listed band.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: 0.3333333333333333 (2 points).

Numerical target 0.33333333; absolute tolerance 0.00066666667. Accepted units: , .

14. (2 points) Calculate the Rf value for the second listed band.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: 0.5555555555555556 (2 points).

Numerical target 0.55555556; absolute tolerance 0.0011111111. Accepted units: , .

15. (2 points) Does a matching ink-band pattern identify one unique pen owner? Explain.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: No. Different pens can contain similar inks, and a class-level match is not a unique personal identification; evaluate additional independent evidence. (2 points).

16. (2 points) Classify the supplied cotton reference fiber as natural or synthetic and state its source category.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: Natural plant fiber. (2 points).

17. (2 points) Explain why a human hair morphology match alone does not identify one person.

answer: Many people can share hair features; morphology generally gives class evidence and must not be treated as unique DNA identification. (2 points).

18. (2 points) A suspect has a powder match but a non-matching fiber and ink. Explain how to report the mixed evidence.

Synthetic complete case 1: crime-scene powder is water-soluble, produces gas with supplied dilute acid, and has the same test pattern as reference sodium bicarbonate. Scene fiber matches the provided cotton reference in microscopy and controlled test observations. A suspect's pen has two chromatography bands at 3 and 5 cm with a solvent front at 9 cm. Reference results are supplied to avoid any need for physical unknowns.

answer: State which findings support and refute an association, discuss transfer and sampling limitations, and avoid turning one class match into a proof of guilt. (2 points).

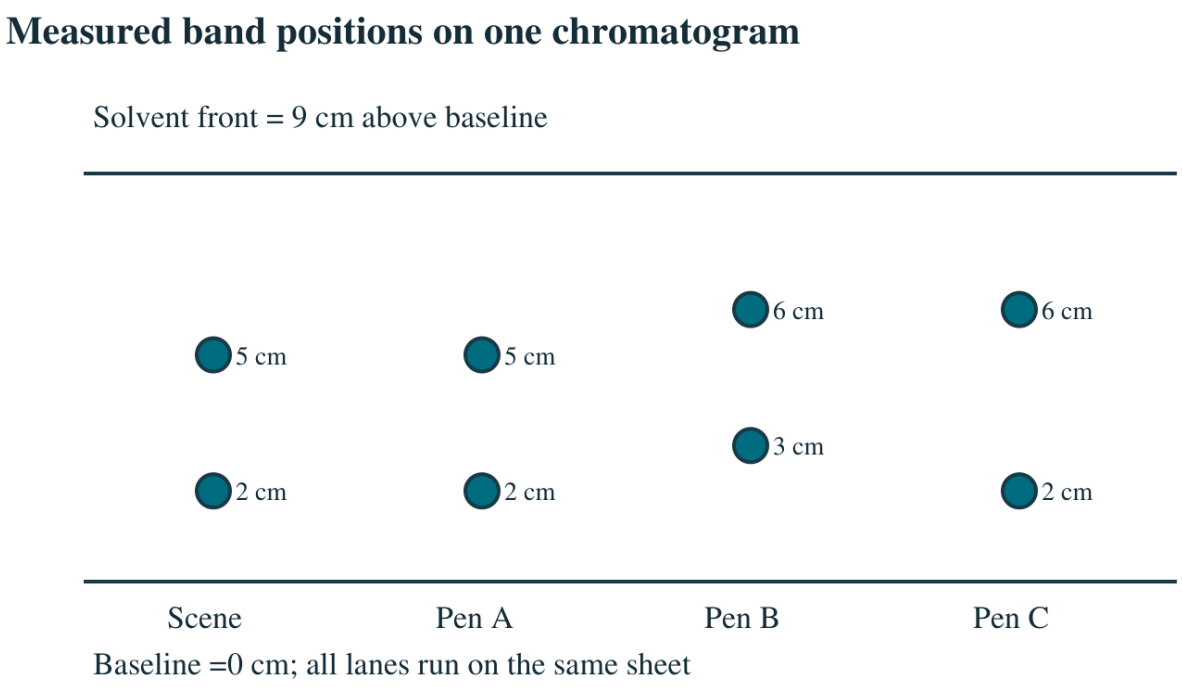
19. (2 points) Describe one control that helps distinguish contamination from a real sample result.

answer: Use a procedural blank or clean reference, separate tools and consistent test conditions, and compare the blank with the sample result. (2 points).

20. (2 points) Why should a complete case report retain contradictory observations rather than omit them?

answer: They constrain hypotheses and prevent selective reporting; a conclusion should reflect the full evidence set and its limitations. (2 points).

21. (2 points) Use Figure F1. Calculate Rf for the lower scene band.



F1. Ink chromatography (original synthetic teaching figure)

answer: 0.22222222 (2 points).

Numerical target 0.22222222; absolute tolerance 0.00066666667. Accepted units: .

22. (2 points) Use Figure F1. Which candidate pen matches both scene band positions?

Refer to F1. Ink chromatography (original synthetic teaching figure) shown with the first question in this figure group.

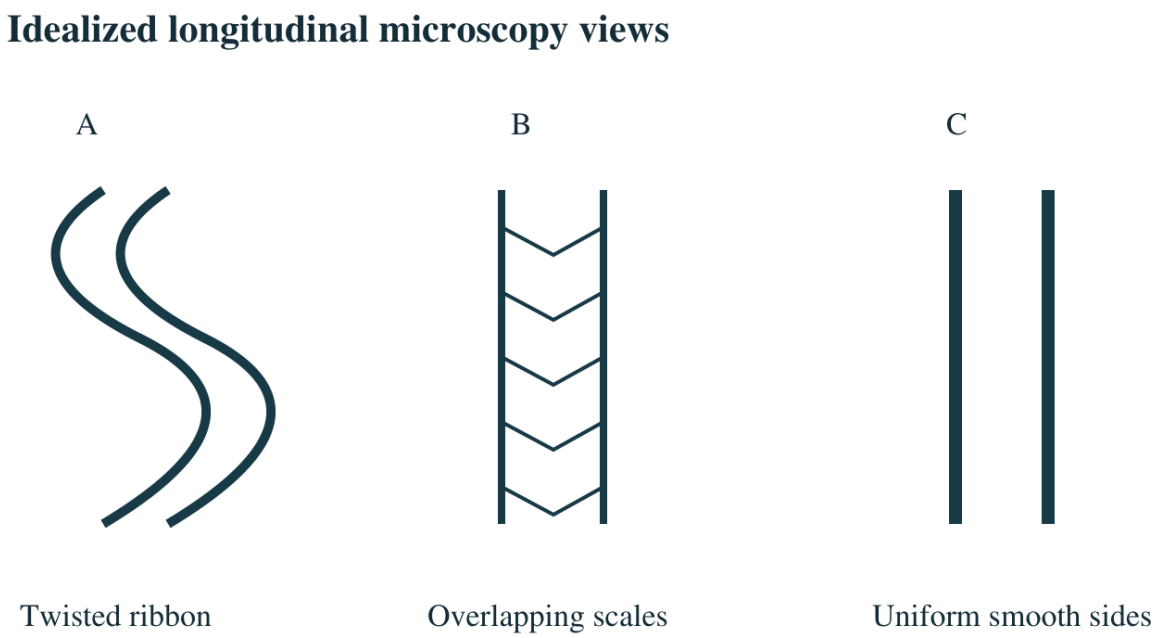
answer: Pen A. (2 points).

23. (2 points) Use Figure F1. Does the match uniquely identify a writer? Explain.

Refer to F1. Ink chromatography (original synthetic teaching figure) shown with the first question in this figure group.

answer: No. Matching ink behavior is class evidence; different pens and owners can share ink formulations, so other independent evidence is required. (2 points).

24. (2 points) Use Figure F2. Which panel is most consistent with the supplied cotton reference morphology?



F2. Fiber morphology sketches (original synthetic teaching figure)

answer: A, the twisted ribbon-like plant fiber. (2 points).

25. (2 points) Use Figure F2. Which panel is most consistent with wool-type cuticle morphology?

Refer to F2. Fiber morphology sketches (original synthetic teaching figure) shown with the first question in this figure group.

answer: B, the overlapping scale pattern. (2 points).

26. (2 points) Use Figure F2. Why is assigning one unique garment from panel C alone unjustified?

Refer to F2. Fiber morphology sketches (original synthetic teaching figure) shown with the first question in this figure group.

answer: Smooth morphology can occur in multiple manufactured fibers and products; additional chemical, physical and comparison evidence is needed for source attribution. (2 points).

27. (2 points) Use Figure F3. Which supplied reference best matches U using the complete pattern?

Matched test conditions: supplied references and unknown

Sample	Water	Dilute acid	Iodine
Ref bicarbonate	Dissolves	Bubbles	No blue-black
Ref cornstarch	Suspension	No bubbles	Blue-black
Unknown U	Dissolves	Bubbles	No blue-black

F3. Powder test station record (original synthetic teaching figure)

answer: Sodium bicarbonate. (2 points).

28. (2 points) Use Figure F3. Which displayed test most directly distinguishes the starch reference through its characteristic color response?

Refer to F3. Powder test station record (original synthetic teaching figure) shown with the first question in this figure group.

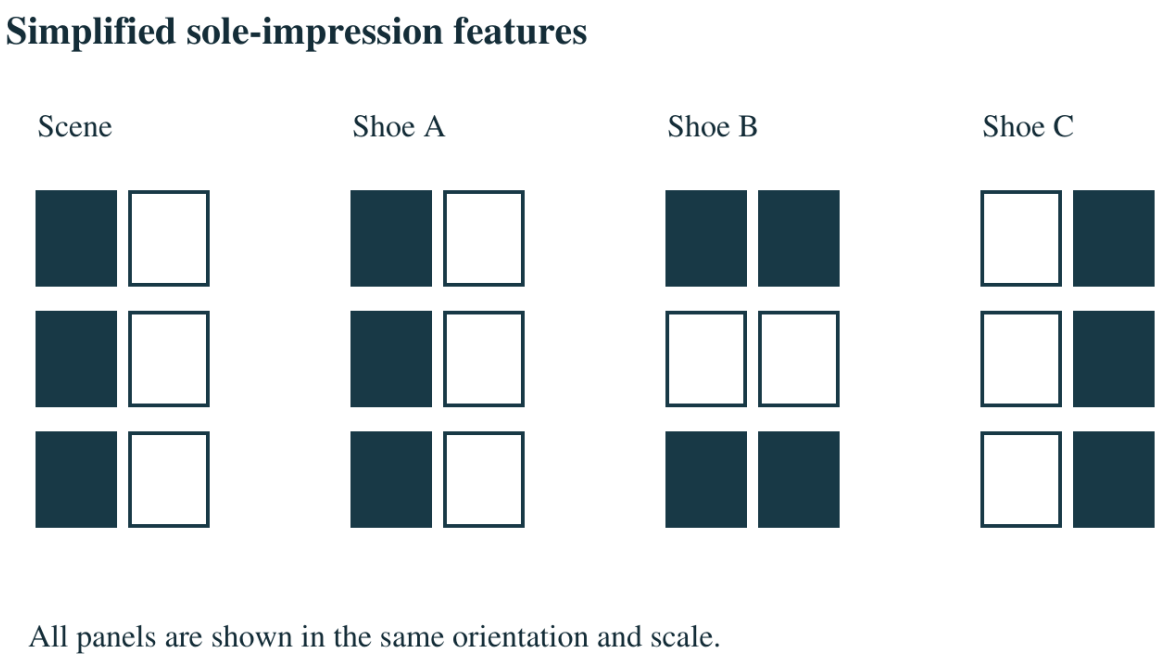
answer: Iodine: the starch reference turns blue-black while U does not. (2 points).

29. (2 points) Use Figure F3. Why should an identification compare the complete row rather than only water solubility?

Refer to F3. Powder test station record (original synthetic teaching figure) shown with the first question in this figure group.

answer: Multiple materials can share solubility; the combined test pattern supplies more discriminating evidence, subject to the available reference set and controls. (2 points).

30. (2 points) Use Figure F4. Which candidate shares the scene class pattern in the displayed orientation?



F4. Class-pattern comparison (original synthetic teaching figure)

answer: Shoe A. (2 points).

31. (2 points) Use Figure F4. How many filled blocks are visible in the scene panel?

Refer to F4. Class-pattern comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: 3 (2 points).

Numerical target 3; absolute tolerance 0.009. Accepted units: .

32. (2 points) Use Figure F4. What additional type of feature could help distinguish shoes sharing this class pattern?

Refer to F4. Class-pattern comparison (original synthetic teaching figure) shown with the first question in this figure group.

answer: Reproducible individual wear, cuts or damage, together with adequate sampling and comparison quality; the generic class pattern alone is not unique. (2 points).